

Rajalakshmi Engineering College

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NeoColab_REC_CS23231_DATA STRUCTURES

REC_DS using C_Week 6_COD_Question 4

Attempt : 1
Total Mark : 10
Marks Obtained : 10

Section 1 : Coding

1. Problem Statement

Kavya, a software developer, is analyzing data trends. She has a list of integers and wants to identify the n th largest number in the list after sorting the array using QuickSort.

To optimize performance, Kavya is required to use QuickSort to sort the list before finding the n th largest number.

Input Format

The first line of input consists of an integer n , representing the size of the array.

The second line consists of n space-separated integers, representing the elements of the array `nums`.

The third line consists of an integer k , representing the position of the largest

number you need to print after sorting the array.

Output Format

The output prints the k-th largest number in the sorted array (sorted in ascending order).

Refer to the sample output for formatting specifications.

Sample Test Case

Input: 6
-1 0 1 2 -1 -4
3

Output: 0

Answer

```
#include <stdio.h>
#include <stdlib.h>

// You are using GCC
int partition(int arr[], int low, int high) {
    int pivot=arr[high];
    int i=low-1;
    for(int j=low;j<=high-1;j++){
        if(arr[j]<pivot){
            i++;
            int temp=arr[i];
            arr[i]=arr[j];
            arr[j]=temp;
        }
    }
    int temp=arr[i+1];
    arr[i+1]=arr[high];
    arr[high]=temp;
    return i+1;
}

void quickSort(int arr[], int low, int high) {
    if(low<high){
        int pi=partition(arr,low,high);
        quickSort(arr,low,pi-1);
```

```
        quickSort(arr,pi+1,high);
    }
}

void findNthLargest(int* nums, int n, int k) {
    quickSort(nums,0,n-1);
    printf("%d\n",nums[n-k]);
}

int main() {
    int n, k;
    scanf("%d", &n);
    int* nums = (int*)malloc(n * sizeof(int));
    for (int i = 0; i < n; i++) {
        scanf("%d", &nums[i]);
    }
    scanf("%d", &k);
    findNthLargest(nums, n, k);
    free(nums);
    return 0;
}
```

Status : Correct

Marks : 10/10